

Set Topology

Lecture 08: Topology Lecture 08 (Supplemental): A Continuous, Bijective Function without a Continuous Inverse

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April 4, 2026

Introduction

In the main lecture on homeomorphisms, we defined a **homeomorphism** as a continuous function that has a continuous inverse. This raises a natural question: Isn't a continuous bijection (a continuous, one-to-one, and onto function) enough? In other areas of mathematics, like linear algebra or group theory, proving a homomorphism is a bijection is sufficient to show it's an isomorphism; the inverse map automatically inherits the required structure.

Topology is different. The condition that the inverse must also be continuous is essential and not redundant. This supplemental lecture provides a classic example of a function that is continuous and bijective, but whose inverse is not continuous, proving it is not a homeomorphism.

Think of topology as the study of “rubber-sheet” geometry — we care about which sets are open, not about distances or angles. A homeomorphism is the topological notion of *isomorphism*: two spaces are homeomorphic if one can be continuously deformed into the other *and back*. The “and back” part is crucial, and this lecture shows exactly why it cannot be dropped.

💡 Summary of Key Ideas

Introduction **Key idea.** A homeomorphism $f : X \rightarrow Y$ requires:

1. f is continuous,
2. f is bijective, and
3. f^{-1} is continuous.

Conditions (1)–(2) alone are *not* sufficient. The rest of these notes construct an explicit counterexample.

1 Prerequisite: The Subspace Topology

To understand the example, we first need to define how to give a topology to a subset of an existing topological space.

If you already have a notion of “open set” for a big space X , you can *restrict* it to any subset $S \subseteq X$ for free: just intersect every open set of X with S . The surprising consequence is that a set that looks “half-open” in X can be genuinely open when viewed inside S .

Definition 1.1 (Subspace Topology). Let (X, τ) be a topological space, and let $S \subseteq X$. The **subspace topology** on S is the collection

$$\tau_S = \{U \cap S \mid U \in \tau\}.$$

In simple terms, a set is considered open in S if it is the intersection of an open set in X with S .

Step 1 — Claim. τ_S as defined above actually satisfies all the axioms of a topology on S .

Step 2 — Empty set and S itself. $\emptyset = \emptyset \cap S \in \tau_S$ (since $\emptyset \in \tau$) and $S = X \cap S \in \tau_S$ (since $X \in \tau$).

Step 3 — Arbitrary unions. If $\{U_\alpha \cap S\}_\alpha \subseteq \tau_S$, then $\bigcup_\alpha (U_\alpha \cap S) = (\bigcup_\alpha U_\alpha) \cap S$. Since $\bigcup_\alpha U_\alpha \in \tau$, the union is in τ_S .

Step 4 — Finite intersections. $(U_1 \cap S) \cap (U_2 \cap S) = (U_1 \cap U_2) \cap S \in \tau_S$ since $U_1 \cap U_2 \in \tau$.

Conclusion. τ_S satisfies all three topology axioms. $\square$

Examples of the Subspace Topology

Example 1.1 (Open interval inside $\mathbb{R}$). Let $X = \mathbb{R}$ (standard topology) and $S = (0, 2)$. The set $(0.5, 1.5)$ is open in S because

$$(0.5, 1.5) = (0.5, 1.5) \cap (0, 2),$$

and $(0.5, 1.5)$ is already open in $\mathbb{R}$. In fact, every open set of $\mathbb{R}$ contained in $(0, 2)$ is also open in the subspace $(0, 2)$.

Example 1.2 (The half-open interval $[0, 1)$). Let $X = \mathbb{R}$ with its standard topology (where open sets are unions of open intervals) and let $S = [0, 1)$.

- The set $(0.2, 0.5)$ is open in $[0, 1)$ because it is the intersection of the open set $(0.2, 0.5)$ from $\mathbb{R}$ with $[0, 1)$.
- The set $[0, 0.5)$ is also open in $[0, 1)$. This might seem strange, but it follows the definition. Take $U = (-0.1, 0.5)$, which is open in $\mathbb{R}$. Then

$$U \cap S = (-0.1, 0.5) \cap [0, 1) = [0, 0.5),$$

so $[0, 0.5)$ is open in the subspace $[0, 1)$.

This situation is illustrated in Figure 1.

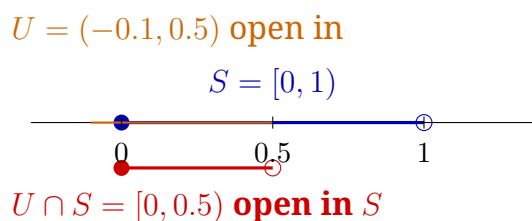


Figure 1: The half-open set $[0, 0.5)$ is open in the subspace $[0, 1)$ because it equals $(-0.1, 0.5) \cap [0, 1)$.

Example 1.3 (The unit circle). Let $X = \mathbb{R}^2$ with its standard topology and $S = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$.

An open set in S is the intersection of an open set in $\mathbb{R}^2$ with S . For instance, intersecting an open ball (an open set in $\mathbb{R}^2$) with the circle gives an open arc. This is illustrated in Figure 2.

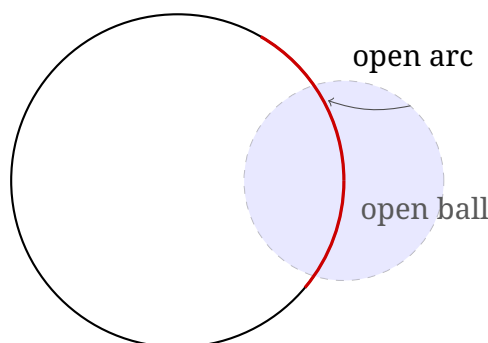


Figure 2: An open ball in $\mathbb{R}^2$ intersects the circle S to form an open arc, which is open in the subspace topology of S .

Mistake 1. Students often say “ $[0, 0.5)$ cannot be open because it has a closed left endpoint.” Openness is *relative to the ambient space*. In $\mathbb{R}$, the set $[0, 0.5)$ is not open. But in the subspace $[0, 1)$, it is open because it can be expressed as $(\text{open set in } \mathbb{R}) \cap [0, 1)$.

Mistake 2. Confusing τ_S with the topology τ restricted to sets *contained in* S . The subspace topology contains $U \cap S$ for *every* $U \in \tau$, even when U is not a subset of S .

💡 Summary of Key Ideas

Subspace Topology Definition. For (X, τ) and $S \subseteq X$: $\tau_S = \{U \cap S \mid U \in \tau\}$.

Key fact. A set $V \subseteq S$ is open in S iff there exists an open set U in X with $V = U \cap S$.

Surprising consequence. Half-open sets like $[0, a)$ are open in $[0, 1)$ even though they are not open in $\mathbb{R}$.

2 The Counterexample: Wrapping an Interval onto a Circle

We will now construct a continuous bijection from the half-open interval $X = [0, 1)$ to the unit circle Y whose inverse is not continuous.

2.1 The Spaces

- **Domain X :** The interval $[0, 1)$ with the subspace topology inherited from $\mathbb{R}$.
- **Codomain Y :** The unit circle embedded in the complex plane (or $\mathbb{R}^2$), with the subspace topology from $\mathbb{R}^2$. Recall $Y = \{z \in \mathbb{C} \mid |z| = 1\}$.

2.2 The Function

Think of bending a rubber rod $[0, 1)$ into a circle. You glue the left endpoint 0 to the position $(1, 0)$ on the circle, then wind counter-clockwise. The right end of the rod approaches $(1, 0)$ but never reaches it — after all, 1 is not in the domain. The rod covers the entire circle exactly once, which is why φ is bijective.

Define $\varphi : [0, 1) \rightarrow Y$ by

$$\varphi(x) = e^{i2\pi x}.$$

Using Euler's formula, this can be written in coordinates as

$$\varphi(x) = (\cos(2\pi x), \sin(2\pi x)).$$

Key values.

$$\varphi(0) = (1, 0), \quad \varphi(0.25) = (0, 1), \quad \varphi(0.5) = (-1, 0), \quad \varphi(0.75) = (0, -1),$$

and as $x \rightarrow 1^-$, $\varphi(x) \rightarrow (1, 0)$ from the fourth-quadrant side. The wrapping is illustrated in Figure 3.

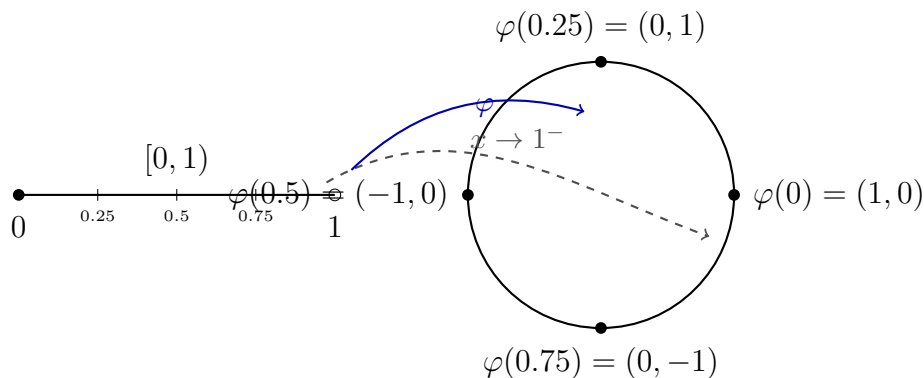


Figure 3: The map $\varphi(x) = e^{2\pi i x}$ wraps $[0, 1)$ counter-clockwise around Y . The right end of the interval approaches $(1, 0)$ but never arrives.

Example 2.1 (Computing values of φ). Compute $\varphi(1/6)$ and $\varphi(5/6)$ and locate them on the unit circle.

$$\begin{aligned}\varphi\left(\frac{1}{6}\right) &= \left(\cos \frac{\pi}{3}, \sin \frac{\pi}{3}\right) = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right), \\ \varphi\left(\frac{5}{6}\right) &= \left(\cos \frac{5\pi}{3}, \sin \frac{5\pi}{3}\right) = \left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right).\end{aligned}$$

These are the standard 60 and 300 positions on the unit circle.

3 Analysis of the Function and its Inverse

3.1 φ is Continuous

φ is built from cosine and sine, which are continuous everywhere. A function into $\mathbb{R}^2$ is continuous precisely when both its coordinate functions are continuous — so continuity of φ is immediate.

The component functions $\cos(2\pi x)$ and $\sin(2\pi x)$ are well-known continuous functions from $\mathbb{R}$ to $\mathbb{R}$. A function into $\mathbb{R}^2$ is continuous if and only if its component functions are continuous. Therefore, φ is continuous.

Step 1 — Goal. Show $\varphi : [0, 1) \rightarrow \mathbb{R}^2$ is continuous.

Step 2 — Decompose. Write $\varphi(x) = (f_1(x), f_2(x))$ where $f_1(x) = \cos(2\pi x)$ and $f_2(x) = \sin(2\pi x)$.

Step 3 — Apply the product criterion. A map into $\mathbb{R}^2$ is continuous iff each coordinate function is continuous. Both f_1 and f_2 are compositions of the continuous linear map $x \mapsto 2\pi x$ with the continuous functions cosine and sine. Compositions of continuous maps are continuous.

Conclusion. φ is continuous. □

Example 3.1 (Continuity at $x = 0$). We verify directly that φ is continuous at $x = 0$. Let $\varepsilon > 0$. We need to find $\delta > 0$ such that whenever $x \in [0, 1)$ and $|x - 0| < \delta$, we have $\|\varphi(x) - \varphi(0)\| < \varepsilon$.

$$\|\varphi(x) - (1, 0)\| = \sqrt{(\cos 2\pi x - 1)^2 + \sin^2 2\pi x} = \sqrt{2 - 2\cos 2\pi x} = 2|\sin \pi x|.$$

Since $|\sin \pi x| \leq \pi|x|$ for small $|x|$, choosing $\delta = \varepsilon/(2\pi)$ gives $\|\varphi(x) - (1, 0)\| \leq 2\pi|x| < \varepsilon$.

3.2 φ is Bijective (One-to-One and Onto)

Injectivity: as x travels from 0 to just below 1, the point $e^{2\pi i x}$ makes exactly one full loop. No two distinct inputs land on the same output. **Surjectivity:** every angle on the circle corresponds to some $x \in [0, 1)$ via $x = \theta/(2\pi)$.

Injectivity (One-to-one): Assume $\varphi(x_1) = \varphi(x_2)$. Then $e^{i2\pi x_1} = e^{i2\pi x_2}$. This implies the exponents differ by an integer multiple of $2\pi i$:

$$i2\pi x_1 = i2\pi x_2 + i2\pi k \quad \text{for some } k \in \mathbb{Z}.$$

Simplifying gives $x_1 - x_2 = k$. Since $x_1, x_2 \in [0, 1)$, their difference lies in $(-1, 1)$. The only integer in this interval is $k = 0$, hence $x_1 = x_2$. Thus φ is injective.

Surjectivity (Onto): Any point on the unit circle can be written as $e^{i\theta}$ for some $\theta \in [0, 2\pi)$. Choose $x = \theta/(2\pi) \in [0, 1)$. Then $\varphi(x) = e^{i\theta}$. Hence every point of is attained, so φ is surjective.

Injectivity — Goal. $\varphi(x_1) = \varphi(x_2) \Rightarrow x_1 = x_2$.

Step 1. $e^{2\pi i x_1} = e^{2\pi i x_2}$ implies $e^{2\pi i(x_1 - x_2)} = 1$.

Step 2. The exponential $e^{2\pi i t} = 1$ iff $t \in \mathbb{Z}$. So $x_1 - x_2 = k$ for some $k \in \mathbb{Z}$.

Step 3. Both $x_1, x_2 \in [0, 1)$, so $-1 < x_1 - x_2 < 1$. The only integer in $(-1, 1)$ is 0.

Conclusion. $k = 0$, so $x_1 = x_2$. □

Example 3.2 (Checking injectivity concretely). Suppose $\varphi(x_1) = \varphi(x_2)$ with $x_1 = 0.1$ and suppose we want to check what x_2 could be. We need $\cos(2\pi \cdot 0.1) = \cos(2\pi x_2)$ and $\sin(2\pi \cdot 0.1) = \sin(2\pi x_2)$. The only solutions to both equations simultaneously are $x_2 = 0.1 + k$ for integer k . The only value in $[0, 1)$ is $x_2 = 0.1$. So $x_1 = x_2$, confirming injectivity.

Since φ is a continuous bijection, it has a set-theoretic inverse $\varphi^{-1} : \rightarrow [0, 1)$.

3.3 φ^{-1} is NOT Continuous

φ^{-1} “unwraps” the circle back to the interval. Everything is fine except at the point $(1, 0)$: points just clockwise of $(1, 0)$ on the circle (fourth quadrant) map to values close to 1 in $[0, 1)$, while $(1, 0)$ itself maps to 0. This jump is the discontinuity. See Figure 4.

To prove the inverse is not continuous, we must find an open set in the codomain $[0, 1)$ whose preimage under φ^{-1} is not open in the domain.

1. **Choose an open set in $[0, 1)$.** Let $U = [0, 0.5)$. As established in Section 2, U is open in the subspace topology of $[0, 1)$.
2. **Find the preimage under φ^{-1} .** The preimage of U under φ^{-1} is $(\varphi^{-1})^{-1}(U) = \varphi(U)$.

$$\varphi(U) = \varphi([0, 0.5)).$$
3. **Analyze $\varphi(U)$ in .** $\varphi(U)$ consists of points on the unit circle obtained by wrapping $[0, 0.5)$. This is the upper semicircle including the point $(1, 0)$ but not including $(-1, 0)$ (since 0.5 is not included).
4. **Is $\varphi(U)$ open in ?** Consider the point $p = (1, 0) \in \varphi(U)$. For $\varphi(U)$ to be open in the subspace topology of , there must exist an open arc around p that is entirely contained in $\varphi(U)$. However, any open arc around $(1, 0)$ contains points just “above” it (in the first quadrant) and points just “below” it (in the fourth quadrant). Points in the fourth quadrant are not in $\varphi(U)$ (they would correspond to angles in $(3\pi/2, 2\pi)$, i.e., $x \in (0.75, 1)$). Therefore, no open neighborhood of p on the circle is fully contained in $\varphi(U)$. Hence $\varphi(U)$ is not open in .

Goal. Show $\varphi^{-1} : \rightarrow [0, 1)$ is not continuous.

Step 1. It suffices to find an open set $V \subseteq [0, 1)$ such that $(\varphi^{-1})^{-1}(V)$ is not open in .

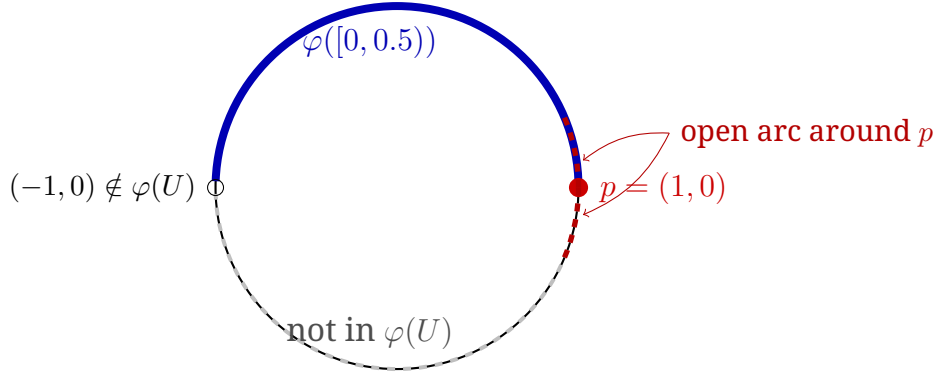


Figure 4: The image $\varphi([0, 0.5))$ is the upper semicircle (thick blue arc) including $(1, 0)$. Every open arc around $p = (1, 0)$ dips into the lower (unshaded) part of the circle, so $\varphi([0, 0.5))$ is not open in $\mathbb{C}$. This is the source of the discontinuity of φ^{-1} .

Step 2. Choose $V = [0, 0.5)$. This is open in $[0, 1)$ (shown in Section 2, Example 1.2).

Step 3. Compute the preimage: $(\varphi^{-1})^{-1}(V) = \varphi(V) = \varphi([0, 0.5))$, which is the upper semicircle plus the point $(1, 0)$.

Step 4. A set is open in $\mathbb{C}$ iff every point has an open arc neighbourhood inside the set. The point $p = (1, 0)$ is in $\varphi([0, 0.5))$. But any open arc containing p also contains points in the fourth quadrant, which are *not* in $\varphi([0, 0.5))$.

Conclusion. $\varphi([0, 0.5))$ is not open in $\mathbb{C}$, so φ^{-1} is not continuous. $\square$

Example 3.3 (Seeing the discontinuity via sequences). A function is continuous iff it preserves convergent sequences. Consider the sequence of points on :

$$z_n = e^{-2\pi i/n} = (\cos(2\pi/n), -\sin(2\pi/n)).$$

As $n \rightarrow \infty$, $z_n \rightarrow (1, 0)$ in $\mathbb{C}$. Applying φ^{-1} :

$$\varphi^{-1}(z_n) = 1 - \frac{1}{n} \rightarrow 1,$$

but $1 \notin [0, 1)$, so this sequence in $[0, 1)$ has no limit within the domain. Meanwhile $\varphi^{-1}((1, 0)) = 0$. The sequence of outputs $1 - 1/n$ does not converge to 0, confirming the discontinuity of φ^{-1} at $(1, 0)$.

Mistake 1. “ φ^{-1} exists as a set-theoretic function, so it must be continuous.” False. The existence of a bijective inverse says nothing about continuity of that inverse.

Mistake 2. Confusing the preimage $(\varphi^{-1})^{-1}(U)$ with the image $\varphi^{-1}(U)$. By the definition of inverse functions, $(\varphi^{-1})^{-1}(U) = \varphi(U)$ — the image of U under the *original* map φ . This is the identity that makes the argument work.

💡 Summary of Key Ideas

Analysis of φ **Continuity of φ** . Immediate from $\cos$ and $\sin$ being continuous and the product criterion for continuity into $\mathbb{R}^2$.

Injectivity. $\varphi(x_1) = \varphi(x_2)$ forces $x_1 - x_2 \in \mathbb{Z}$; since both values lie in $[0, 1)$, the only possibility is $x_1 = x_2$.

Surjectivity. Every $e^{i\theta}$ is hit by $x = \theta/(2\pi) \in [0, 1)$.

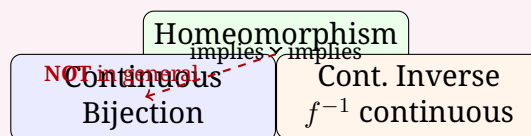
Discontinuity of φ^{-1} . The set $[0, 0.5)$ is open in $[0, 1)$, but its image $\varphi([0, 0.5))$ under φ is *not* open in $\mathbb{C}$ (it fails at $(1, 0)$).

4 Conclusion

We have successfully constructed a continuous and bijective function $\varphi : [0, 1) \rightarrow \mathbb{C}$ whose inverse is not continuous. This demonstrates that φ is *not* a homeomorphism, and it proves that the requirement for a continuous inverse is a necessary and non-negotiable part of the definition of a homeomorphism. Simply being a continuous bijection is not enough to preserve the topological structure between spaces.

💡 Summary of Key Ideas

The Big Picture



The example $\varphi : [0, 1) \rightarrow \mathbb{C}$ is a continuous bijection that is *not* a homeomorphism. Adding the hypothesis that X is compact (or Y is Hausdorff) would close the gap — but without it, no shortcut exists.

Practice Problems

1. Consider the same function $\varphi(x) = e^{i2\pi x}$, but change the domain to the open interval $X = (0, 1)$. What is the image $\varphi(X)$? Is this new function $\varphi : (0, 1) \rightarrow \varphi(X)$ (with $\varphi(X)$ having the subspace topology) a homeomorphism? Justify your answer.
2. Let X be the set with the discrete topology (every subset is open). Let Y be the set with the standard topology. Consider the identity function $\text{id} : X \rightarrow Y$ defined by $\text{id}(x) = x$.
 - (a) Show that id is a continuous bijection.
 - (b) Show that id is not a homeomorphism by proving that its inverse is not continuous.

3. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = x^3$. Both the domain and codomain have the standard topology. Prove that f is a homeomorphism.
4. **(Additional)** Let $g : [0, 1) \rightarrow [0, 1)$ be defined by $g(x) = x/2$. Is g a homeomorphism? What about $h : (0, 1) \rightarrow (0, 1)$ defined by $h(x) = x/2$? (Hint: think about whether the image is all of $(0, 1)$.)
5. **(Additional, Conceptual)** Suppose $f : X \rightarrow Y$ is a continuous bijection and X is a *compact* topological space while Y is *Hausdorff*. Prove that f is a homeomorphism. (This shows the counterexample φ must fail compactness — and indeed $[0, 1)$ is not compact.)

Reference for Further Reading

Munkres, James R. *Topology*. 2nd ed., Pearson, 2000. Chapter 2, Section 22: “The Subspace Topology”.